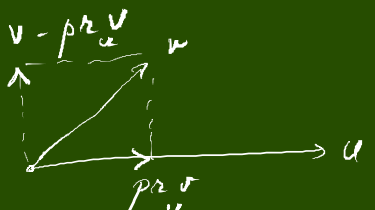
- $(V, \langle , \rangle)$ $\quad u, v \in V, \quad u \neq 0$

$\| pr_u v \| \leq \| v \|$

$pr_u v = \dfrac{\langle v, u \rangle}{\langle u, u \rangle} \cdot u$

$\neq 0 \ (u \neq 0)$



$pr_u v = \alpha \cdot u \qquad \langle v - \alpha u, u \rangle = 0 \Rightarrow \alpha = \dots$

- Bază ortonormată $\quad q_1 \dots q_n \in V$

$$\langle q_i, q_j \rangle = \begin{cases} 0, & i \neq j \\ 1, & i = j \end{cases}$$

- Bază $\{v_1 \dots v_n\}$

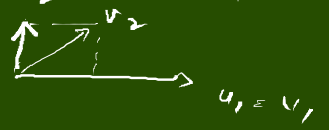
↓

bază ortogonală $\{u_1 \dots u_n\}$

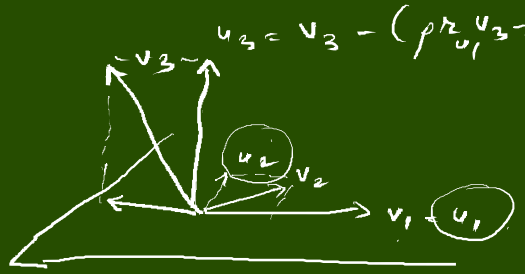
↓

bază ortonormată $\{q_1 \dots q_n\}$

$u_2 = v_2 - proj_{u_1} v_2$



$u_3 = v_3 - (proj_{u_1} v_3 + proj_{u_2} v_3)$



---

esempio $\mathbb{R}^3$, $v_1 = \begin{pmatrix} 2 \\ 1 \\ 2 \end{pmatrix}$, $v_2 = \begin{pmatrix} 6 \\ -3 \\ 0 \end{pmatrix}$, $v_3 = \begin{pmatrix} 3 \\ -3 \\ 3 \end{pmatrix}$

$\{v_1, v_2, v_3\}$ base di $\mathbb{R}^3$

$\langle v_1, v_2 \rangle = 2 \cdot 6 + 1 \cdot (-3) + 2 \cdot 0 = 9 \neq 0$

$v_1 \not\perp v_2$

$\langle v_1, v_3 \rangle = 9 \neq 0$

$\langle v_2, v_3 \rangle = 27 \neq 0$

$u_1 = v_1 = \begin{pmatrix} 2 \\ 1 \\ 2 \end{pmatrix}$

$u_2 = v_2 - \operatorname{pr}_{u_1} v_2 = v_2 - \dfrac{\langle v_2, u_1 \rangle}{\langle u_1, u_1 \rangle} u_1$

$= \begin{pmatrix} 6 \\ -3 \\ 0 \end{pmatrix} - \dfrac{9}{9} \begin{pmatrix} 2 \\ 1 \\ 2 \end{pmatrix} = \begin{pmatrix} 4 \\ -4 \\ -2 \end{pmatrix}$

$u_3 = v_3 - \operatorname{pr}_{u_1} v_3 - \operatorname{pr}_{u_2} v_3 =$

$= v_3 - \dfrac{\langle v_3, u_1 \rangle}{\langle u_1, u_1 \rangle} u_1 - \dfrac{\langle v_3, u_2 \rangle}{\langle u_2, u_2 \rangle} u_2$

$= \begin{pmatrix} 3 \\ -3 \\ 3 \end{pmatrix} - \dfrac{9}{9} \begin{pmatrix} 2 \\ 1 \\ 2 \end{pmatrix} - \dfrac{18}{36} \begin{pmatrix} 4 \\ -4 \\ -2 \end{pmatrix}$

$= \begin{pmatrix} -1 \\ -2 \\ 2 \end{pmatrix}$

$\langle u_1, u_2 \rangle = 0$ ✓ $\quad \langle u_2, u_3 \rangle = 0$ ✓

$\langle u_1, u_3 \rangle = 0$ ✓

$\Rightarrow \{u_1, u_2, u_3\}$ base ortogonal de $\mathbb{R}^3$

$$q_1 = \frac{1}{\|u_1\|} u_1 = \frac{1}{\sqrt{2^2+1^2+2^2}} u_1 = \begin{pmatrix} 2/3 \\ 1/3 \\ 2/3 \end{pmatrix}$$

$$q_2 = \frac{1}{\|u_2\|} u_2 = \frac{1}{6} u_2 = \begin{pmatrix} 2/3 \\ -2/3 \\ -1/3 \end{pmatrix}$$

$$q_3 = \frac{1}{\|u_3\|} u_3 = \frac{1}{3} u_3 = \begin{pmatrix} -1/3 \\ -2/3 \\ 2/3 \end{pmatrix}$$

$\Rightarrow B = \{q_1, q_2, q_3\}$ base ortonormada de $\mathbb{R}^3$

$$v = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} \qquad [v]_B = ? \begin{pmatrix} \lambda_1 \\ \lambda_2 \\ \lambda_3 \end{pmatrix}$$

$$\boxed{v = \lambda_1 q_1 + \lambda_2 q_2 + \lambda_3 q_3}$$

$$= \langle v, q_1 \rangle q_1 + \langle v, q_2 \rangle q_2 + \langle v, q_3 \rangle q_3$$

$\langle v, g_1 \rangle = \frac{2}{3}$ $\langle v, g_2 \rangle = \frac{2}{3}$ $\langle v, g_3 \rangle = -\frac{1}{3}$

$v = \frac{2}{3} g_1 + \frac{2}{3} g_2 - \frac{1}{3} g_3$

$Q \in \mathcal{U}_{m,n}(\mathbb{R})$ $\quad Q^t Q = I_n$
$(n,m)(m,n)$

exemplu : $Q = \begin{pmatrix} 2/3 & 2/3 \\ 1/3 & -2/3 \\ 2/3 & -1/3 \end{pmatrix}$

$Q^t Q = I_2$

$1 \Rightarrow 3 \Leftrightarrow 4$

$\uparrow \swarrow$

2

$1 \Rightarrow 3$ $\quad x, y \in \mathbb{R}^n$

$\langle Qx, Qy \rangle = (Qx)^t Qy =$

$= x^t \underbrace{Q^{-1}Q}_{I_n} y = x^t y = \langle x, y \rangle$

3 $\Rightarrow$ 2 $\quad \langle q_i, q_j \rangle \overset{?}{=} \begin{cases} 0, & i \neq j \\ 1, & i = j \end{cases}$

$Q \longrightarrow q_i$

$Qe_i = q_i$

$Q = \begin{pmatrix} a_{11} & \cdots & a_{1n} \\ \vdots & & \vdots \\ a_{n1} & \cdots & a_{nn} \end{pmatrix} \quad e_1 = \begin{pmatrix} 1 \\ 0 \\ \vdots \\ 0 \end{pmatrix}$

$Q \cdot e_1 = \begin{pmatrix} a_{11} \\ \vdots \\ a_{n1} \end{pmatrix} = q_1$

$\langle q_i, q_j \rangle = \langle Qe_i, Qe_j \rangle = \langle e_i, e_j \rangle =$

$= \begin{cases} 0, & i \neq j \\ 1, & i = j \end{cases}$

2 $\Rightarrow$ 1

$Q^t Q = \begin{pmatrix} \text{---} & q_1^t & \text{---} \\ \text{---} & q_2^t & \text{---} \\ & \vdots & \\ \text{---} & q_n^t & \text{---} \end{pmatrix} (q_1 | q_2 | \cdots | q_n) =$

$$= \begin{pmatrix} q_1^t q_1 & q_1^t q_2 & \cdots & q_1^t q_n \\ \cdots & \cdots & \cdots & \cdots \\ \cdots & \cdots & \cdots & q_n^t q_n \end{pmatrix} = \begin{pmatrix} 1 & & 0 \\ & \ddots & \\ 0 & & 1 \end{pmatrix} = I_n$$

$$q_i^t q_j = \langle q_i, q_j \rangle = \begin{cases} 0, & i \neq j \\ 1, & i = j \end{cases}$$

$3 \Rightarrow 4 \qquad \|x\| = \sqrt{\langle x, x \rangle}$

$4 \Rightarrow 3 \qquad \langle x, y \rangle = \frac{1}{2}\left(\|x+y\|^2 - \|x\|^2 - \|y\|^2 \right)$

example

$$Q = \begin{pmatrix} 2/3 & 2/3 & -1/3 \\ 1/3 & -2/3 & -2/3 \\ 2/3 & -1/3 & 2/3 \end{pmatrix}$$ matrice ortogonală

$$\Rightarrow Q^t = \begin{pmatrix} 2/3 & 1/3 & 2/3 \\ 2/3 & -2/3 & -1/3 \\ -1/3 & -2/3 & 2/3 \end{pmatrix}$$ ortogonală

1) $A = P \cdot D \cdot P^{-1}$

2) $A_{m \geq n} = Q \begin{pmatrix} R \\ 0 \end{pmatrix}$ — R inversable

Q orthogonale

$$Ax = b \Rightarrow QRx = b \mid Q^t$$

$$\Rightarrow \underbrace{Q^t Q}_{I_n} Rx = Q^t b$$

$$\Rightarrow Rx = Q^t b$$

$x = Q^t b$

$A = (v_1 | \dots | v_n)$

$\operatorname{rang} A = n \iff \{v_1 \dots v_n\}$ lin. indep

$\Rightarrow \{v_1 \dots v_n\}$ base de $\operatorname{Im}(A)$

Alg. Gram - Schmidt pt. det. unei baze ortonormate în $\mathrm{Im}(A) \Rightarrow \{q_1 \dots q_n\}$

$\Rightarrow Q = (q_1 | \dots | q_n)$ matrice ortogonală

$$R = \begin{pmatrix} \langle v_1, q_1 \rangle & \langle v_2, q_1 \rangle & \dots & \langle v_n, q_1 \rangle \\ 0 & \langle v_2, q_2 \rangle & \dots & \langle v_n, q_2 \rangle \\ \vdots & 0 & & \vdots \\ 0 & 0 & \dots & \langle v_n, q_n \rangle \end{pmatrix}$$

$A = QR \qquad \mathrm{Im}(Q) = \mathrm{Im}(A)$

$$A = \begin{pmatrix} 2 & 6 \\ 1 & -3 \\ 2 & 0 \end{pmatrix} \qquad \mathrm{rang}\, A = 2$$

$v_1 \quad v_2$ sîn indep.

$u_1 = v_1 \qquad q_1 = \frac{1}{\|u_1\|} u_1 = \begin{pmatrix} 2/3 \\ 1/3 \\ 2/3 \end{pmatrix}$

$u_2 = v_2 - \mathrm{pr}_{u_1} v_2 \qquad q_2 = \frac{1}{\|u_2\|} u_2 = \begin{pmatrix} 2/3 \\ -2/3 \\ -1/3 \end{pmatrix}$

$$Q = (\xi_1 | \xi_2) = \begin{pmatrix} 2/3 & 2/3 \\ 1/3 & -2/3 \\ 2/3 & -1/3 \end{pmatrix}$$

$$R = \begin{pmatrix} \langle v_1, \xi_1 \rangle & \langle v_2, \xi_1 \rangle \\ 0 & \langle v_2, \xi_2 \rangle \end{pmatrix} = \begin{pmatrix} 3 & 3 \\ 0 & 6 \end{pmatrix}$$

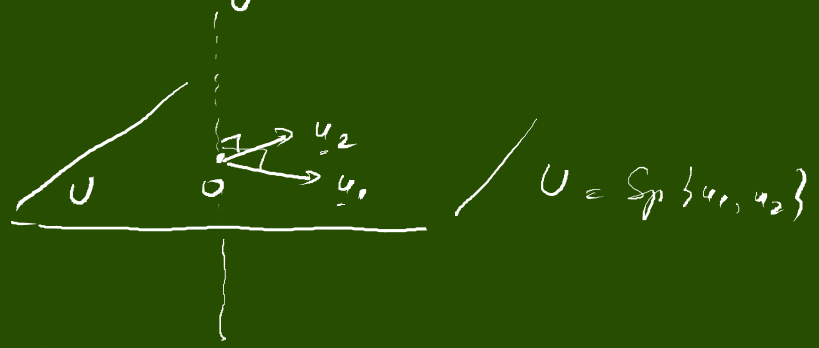
$A = QR$

$(3,2) \quad (3,2)(2,2)$

$$U^{\perp} = \{ v \in V \mid \forall u \in U \ . \ \langle v, u \rangle = 0 \}$$

$v \perp u$

subespaţiu vectorial



$U = Sp\{u_1, u_2\}$

$$U \cap U^{\perp} = \{ 0 \}$$

Jde $v \in U \cap U^\perp$ ⟵ $v \in U$, $v \in U^\perp$

$\|v\|^2 = \langle v, v \rangle = 0 \quad (v \perp v) \Rightarrow v = 0$
$\in U^\perp \in U$

$(U^\perp)^\perp = U$

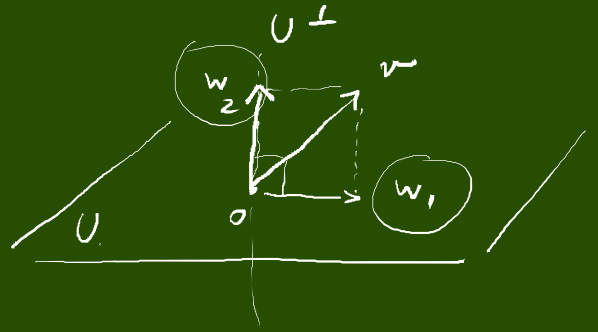
$\supseteq$ "

$\subseteq$ "

Jde $u \in U$, Jde $v \in U^\perp$

$\Rightarrow \langle u, v \rangle = 0 \Rightarrow$

$\Rightarrow u \in (U^\perp)^\perp$



$U = \mathrm{Im}(A) = \mathrm{Sp}\left\{\underbrace{\begin{pmatrix}2\\1\\2\end{pmatrix}}_{v_1}, \underbrace{\begin{pmatrix}6\\-3\\0\end{pmatrix}}_{v_2}\right\} \subseteq \mathbb{R}^3$

$\dim_{\mathbb{R}} U = 2$ (plan care trece prin origine)

$A = \begin{pmatrix}2 & 6\\1 & -3\\2 & 0\end{pmatrix}$

temă : ecuația planului

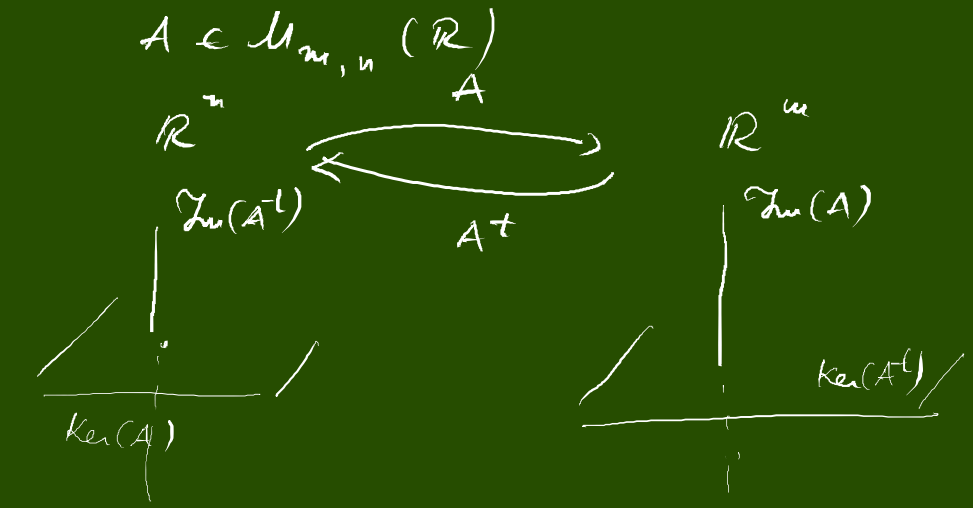
$U^\perp = \{ v = \begin{pmatrix}x\\y\\z\end{pmatrix} \in \mathbb{R}^3 \mid \langle v, u\rangle = 0 \quad \forall u \in U \}$

$= \{ v = \begin{pmatrix}x\\y\\z\end{pmatrix} \in \mathbb{R}^3 \mid \underline{\langle v, v_1\rangle = 0}, \langle v, v_2\rangle = 0 \}$

$\Rightarrow \begin{cases} \underline{2x + y + 2z = 0} \\ \underline{6x - 3y = 0} \end{cases} \quad z = -2x \quad \Rightarrow y = 2x$

$\Rightarrow U^\perp = \ker \underbrace{\begin{pmatrix}2 & 1 & 2\\6 & -3 & 0\end{pmatrix}} = \ker(A^t) = \mathrm{Sp}\left\{\begin{pmatrix}1\\2\\-2\end{pmatrix}\right\}$

$\Rightarrow \begin{cases} \mathrm{Im}(A)^\perp = \ker(A^t) \\ \ker(A)^\perp = \mathrm{Im}(A^t) \end{cases}$



$U \subseteq V$

$v \in V \Rightarrow \exists!\ w_1 \in U$

$w_2 \in U^\perp$

$v = w_1 + w_2$

<u>unicitatea</u> : $v = w_1 + w_2 = z_1 + z_2$

$w_1, z_1 \in U$

$w_2, z_2 \in U^\perp$

$\Rightarrow$ $\underbrace{w_1 - b_1}_{\in U} = \underbrace{b_2 - w_2}_{\in U^\perp}$

$\in U \cap U^\perp = \{0\}$

$\Rightarrow$ $w_1 - b_1 = b_2 - w_2 = 0 \Rightarrow w_1 = b_1$, $w_2 = b_2$

existența — Fie $\{e_1 \dots e_n\}$ bază ortonormată în U

$v = \underbrace{w_1}_{\in U} + \underbrace{w_2}_{\in U^\perp}$

$w_1 = \alpha_1 e_1 + \dots + \alpha_n e_n$

$w_2 = v - w_1 = \underline{v - \alpha_1 e_1 - \dots - \alpha_n e_n}$

$w_2 \perp e_1, \dots, w_2 \perp e_n$ 1

$\Rightarrow 0 = \langle w_2, e_1 \rangle = \langle v, e_1 \rangle - \alpha_1 \langle e_1, e_1 \rangle -$

$- \alpha_2 \underbrace{\langle e_2, e_1 \rangle}_{0} - \dots - \alpha_n \underbrace{\langle e_n, e_1 \rangle}_{0} = \langle v, e_1 \rangle - \alpha_1$

$\Rightarrow \quad \alpha_1 = \langle v, e_1 \rangle$

$0 = \langle w_2, e_2 \rangle \Rightarrow \alpha_2 = \langle v, e_2 \rangle$

$\cdots \cdots \cdots$

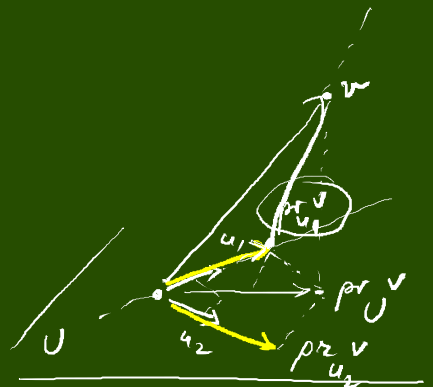
$0 = \langle w_2, e_n \rangle \Rightarrow \alpha_n = \langle v, e_n \rangle$

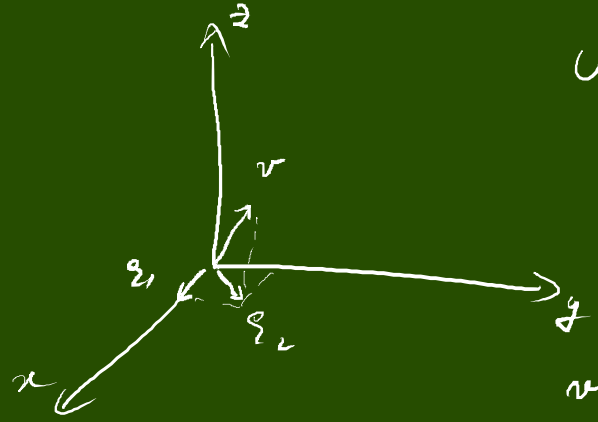
$\Rightarrow w_1 = \underbrace{\frac{\langle v, e_1 \rangle}{\langle e_1, e_1 \rangle} e_1}_{\text{pr}_{e_1} v} + \cdots + \langle v, e_n \rangle e_n \in U$

$w_2 = v - w_1 \in U^\perp$

$v = w_1 + w_2$

$\begin{cases} w_1 = \text{pr}_U v \\ w_2 = \text{pr}_{U^\perp} v \end{cases}$





$$U = \left\{ \begin{pmatrix} x \\ y \\ 0 \end{pmatrix} \mid x, y \in \mathbb{R} \right\}$$

$$\begin{cases} e_1 = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} \\ e_2 = \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix} \end{cases}$$

$$v = \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} \quad \begin{pmatrix} x \\ y \\ z \end{pmatrix}$$

$$pr_U v = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} = e_2 \quad \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}$$

$$\operatorname{pr}_{g_1} v + \operatorname{pr}_{g_2} v = \frac{\langle v, g_1 \rangle}{\langle g_1, g_1 \rangle} g_1 + \frac{\langle v, g_2 \rangle}{\langle g_2, g_2 \rangle} g_2$$

$$= \frac{1}{1} g_1 + \frac{2}{2} g_2 = \begin{pmatrix} 2 \\ 0 \end{pmatrix}$$

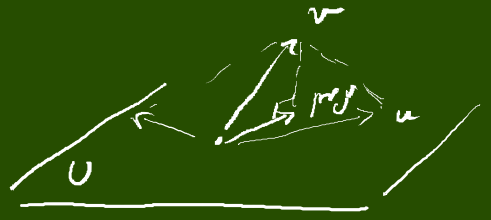
baza $\{g_1, g_2\}$ nu e ortonormată! $\neq \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix} = \operatorname{pr}_U v$

$Q^t Q = id$

$Q Q^{-1} = id$ doar dc Q pătratică

$$\operatorname{pr}_U v = \sum \operatorname{pr}_{g_i} v = \sum \langle v, g_i \rangle g_i$$

$$= Q Q^t v \qquad Q = (g_1 | \dots | g_n)$$



$\| \operatorname{pr}_U v \| \leq \| v \|$